

Assignment 03 – Cognitive Load Analysis

Title: Analyzing Cognitive Principles in Interface Design

Objective: To apply knowledge of cognitive processes to evaluate and improve user interface design

Instructions

1. Choose a Digital Interface

Select a digital interface that you use frequently. Examples include:

- A social media app (e.g., Instagram, Twitter, WeChat, Sina Weibo, Douyin)
- An e-commerce platform (e.g., Amazon, Taobao, Meituan, Pinduoduo)
- A banking or financial app (e.g., Alipay, WeChat Pay, UnionPay)
- A learning management system (e.g., Moodle, Blackboard, Tencent Classroom, Udemy)
- A mobile productivity tool or utility app (Baidu maps, DiDi, Google Docs/Sheets)

2. Analyze Four Cognitive Processes

Identify and analyze **three** of the following cognitive processes that are involved when a user interacts with your chosen interface. For each, do the following:

- 1. Attention:** How does the interface guide the user's attention to important elements (e.g., notifications, calls to action)? Are there distractions (e.g., pop-ups, ads)?
- 2. Memory:** Does the interface depend on the user to **remember information** (recall), or does it help them by presenting recognizable options (recognition)?
- 3. Decision-Making:** How does the interface help users **make informed choices**? Are options clear and easy to compare? Are consequences of actions visible?
- 4. Perception:** How do users visually or audibly **perceive the layout and information**? Are visual cues, icons, fonts, and spacing helping or hurting clarity?

Deliverables

A short written report (approx. **600–700 words**) that includes:

Introduction: Briefly describe the interface you chose.

Four analysis sections (one for each cognitive process), each with:

- 1. Description of the interface's behavior:**

- Explain **how the interface behaves** in relation to the specific cognitive process.
- Describe **what the user sees, hears, or does** while interacting with the interface.

Example (for Attention):

“When the user opens the app, a promotional banner at the top blinks repeatedly, while the main navigation menu remains static at the bottom.”

2. Specific example:

- Provide a **real instance or screenshot** (if included) where this behavior occurs.
- Describe a **task or moment** where the cognitive process is clearly engaged.

Example (for Memory):

“During login, the user must remember their username and a security image chosen during registration, which can be hard after a long break.”

3. Analysis of support/hindrance:

- Evaluate whether the interface **supports or overloads** that cognitive process.
- Consider **usability, mental effort, and user frustration or ease**.
- Use clear reasoning based on principles you've studied.

Example (for Decision-Making):

“The product comparison page presents prices and specs clearly in a table format, which helps reduce the cognitive load of evaluating choices.”

4. Suggested improvement

- Propose a specific design change to **enhance usability** and **reduce cognitive strain**.
- Keep it practical and based on real design strategies (e.g., using icons, limiting options, adding help text, autofill, breadcrumbs).

Example (for Perception):

“Increase contrast between text and background for better readability, especially for visually impaired users.”

Conclusion: Summarize key insights and overall recommendations

Final Deliverable Structure:

Section	Word Count	Purpose
Introduction	~100	Describe the selected interface and why you chose it
4 Analysis Sections	~400–500	120–150 words each section, based on structure above
Conclusion	~100	Reflect on overall design effectiveness and key takeaways

Submission Guidelines:

- Submit your assignment report in a PDF format until midnight on May 15, 2026.
- Name your assignment as a03_id_name.pdf where id is your student id, name is your first name. For example, a03_123456_sophie.pdf
- Submit your assignment through the designated platform.